Huber Group

Regression on Latent Spaces for the Analysis of Multi-Condition Single-Cell RNA-Seq Data

Constantin Ahlmann-Eltze & Wolfgang Huber

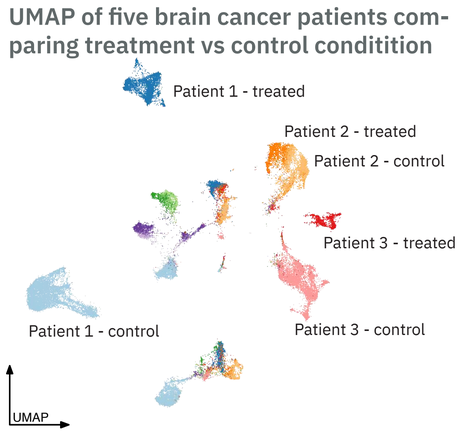
Background

Single-cell RNA-seq data allows researchers to profile the gene expression of 10,000s of cells. Multi-condition single-cell data reveals expression differences between corresponding cell subpopulations in different conditions.

Here, we propose to use regression on latent spaces to simultaneously account for variance from known and latent factors. Our approach is built around multivariate regression on Grassmann manifolds. We use the method to analyze a drug treatment experiment on brain tumor biopsies. The method is a versatile new approach for identifying differentially expressed genes from single-cell data of heterogeneous cell subpopulations under arbitrary experimental designs without clustering.

To foster the adoption of the method, we implemented an R package that is available on github.com/const-ae/lemur.

UMAP of five brain cancer patients comparing treatment vs control condition

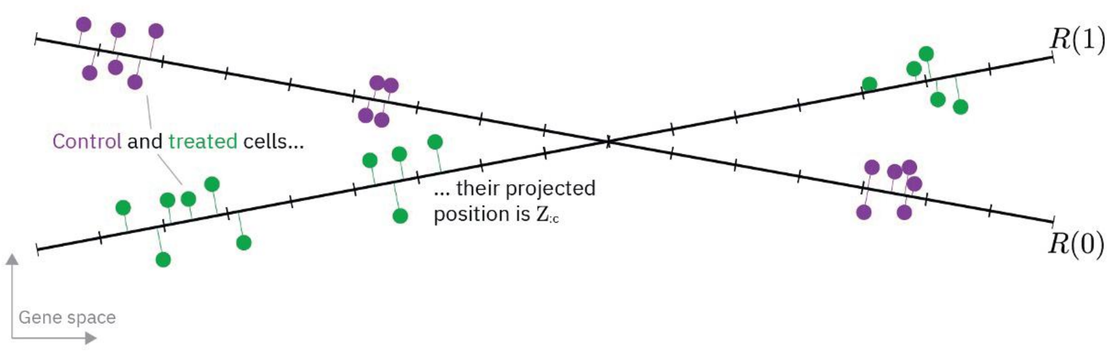


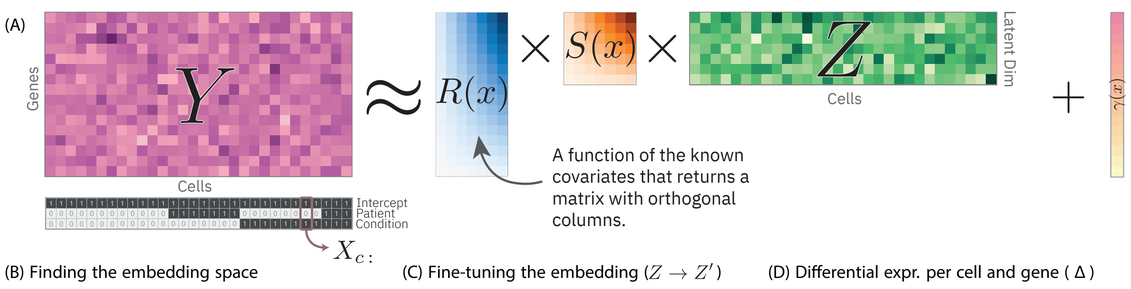
Aim

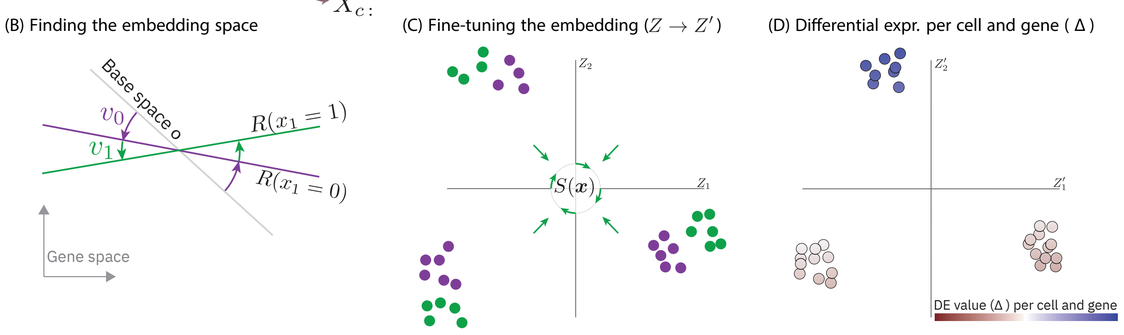
- Cluster and cell type-free integration of experimental conditions
- Support for arbitrary experimental designs
- Differential expression analysis for arbitrary cell populations

Method

Integrating experimental conditions using geodesic regression on the Grassmann manifold







$$f(x_i, z_i) = R(x_i)S(x_i)z_i$$

$$R(x) = \mathrm{Exp}_b(v_0 + x_1 v_1)$$

Not the familiar exp() function!

A Grassmann manifold is the set of all k-dimensional linear subspaces in an n dimensional space. Its elements can be expressed as orthogonal n×k matrices.
The geodesic between two k-planes corresponds to rotating one plane onto the other. Geodesic regression finds the rotation to match a given set of dependent on some known covariates.

We can calculate the differential expression for each gene and cell by contrasting the hypothetical case where we ask what expression level we would expect for a cell at a specific position z_i in the latent space in each condition.

$$\mathrm{diff}_i = f(X^{(\mathrm{WT})}, z_i) - f(X^{(\mathrm{Pert})}, z_i)$$

Previous work

Geodesic regression (Fletcher 2011, Rentmeesters 2011) generalizes linear regression to optimization problems on manifolds, where ***x*** is a vector of known covariates, *b* is a point on the manifold, *v* is a tangent vector, and Exp is the exponential map, and *d* is the norm of the logarithmic map.

$$\arg\min_{b,v} \frac{1}{2}\sum_{i=1}^{N} d(\mathrm{Exp}_b(x_i v), y_i)^2$$

$$\arg\min_{b,v_1,\ldots,v_K} \frac{1}{2}\sum_{i=1}^{N} d\left(\mathrm{Exp}_b\left(\sum_{k=1}^{K} X_{ik} v_k\right), y_i\right)^2$$

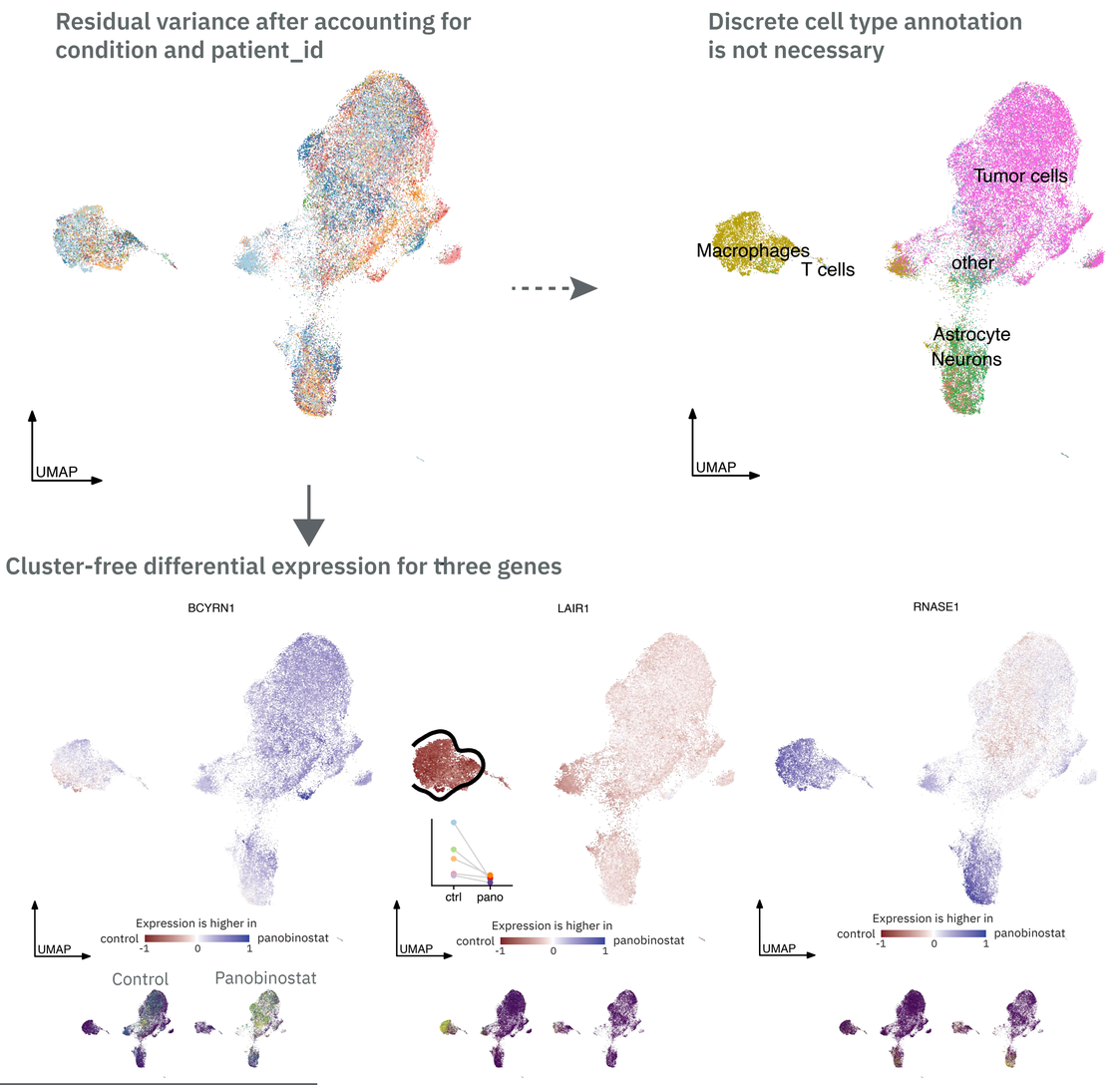
Kim et al. (2014) extended geodesic regression to multivariate settings, where we optimize over a set of design vectors v_k and the covariates are stored in a design matrix *X*.

Optimizing the coefficients vk is difficult. Thus Kim et al. suggested to work in the tangent space of a base point and optimize the values there ($y_i^l = \mathrm{Log}_b(y_i)$)

$$\arg\min_{v_1,\ldots,v_K} \frac{1}{2}\sum_{i=1}^{N} \left|\left| \sum_{k=1}^{K} X_{ik} v_k - y_i^l \right|\right|^2$$

Results

Gene expression changes in brain cancer due to drug treatment



Code

```
library(lemur)

fit <- lemur(sce, design = ~ patient_id + condition, n_embedding = 15)
fit <- align_harmony(fit)
fit <- test_de(fit, contrast = cond(condition = "ctrl") -
                               cond(condition = "panobinostat"))
nei <- find_de_neighborhoods(fit, group_by = vars(patient_id, condition))
```

Conclusion

- Using differential embeddings, we can reverse the usual analysis workflow for multi-condition single-cell data
- After integrating arbitrary conditions, we identify all genes that are differentially expressed in some cell population
- Lastly, we can identify the region where a gene is differentially expressed and then assign a biological function to the set of cells

References

- Deconvolution of cell type-specific drug responses in tumor tissue with single-cell RNA-seq, Zhao (2021)
- Manopt.jl, Ronny Bergmann et al. (2022)
- Multivariate General Linear Models on Riemannian Manifolds with Applications to Statsitical Analysis of Diffusion Weighted Images, Kim et al. (2014)
- Geodesic regression on Riemannian manifolds, Fletcher (2011)

constantin.ahlmann@embl.org @const_ae @const-ae

EMBL Heidelberg
Meyerhofstraße 1 · 69177 Heidelberg · Germany
T +49 6221 3870 · www.embl.org